Option Pricing using Quantum Computers

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Introduction

Options

Options are financial derivatives that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified price, called the strike price (K), before or on a certain date known as the expiration date (T).

Imagine you want to buy a stock currently priced at $50, but you're unsure if its price will go up. You buy a call option with a strike price of $55, expiring in a month, for a small premium. If the stock price rises above $55, you can buy it at this lower price, profiting from the difference. If the stock doesn't reach $55, you only lose the premium you paid for the option. This flexibility makes options useful for speculation and managing risk.

Types of Options

	Single-asset	Multi-asset
Path-independent	European put/call	Basket option
Path-dependent	Barrier & Asian options	Multi-asset barrier options

Table: Example of the different option types.

- **European Put/Call**: Standard options that can only be exercised at expiration. A call allows buying, and a put allows selling the underlying asset at a predetermined strike price.
- **Basket Option**: An option based on a weighted basket of multiple assets, where the payoff depends on the combined performance of these assets.
- **Barrier & Asian Options**: Barrier options are activated or nullified when the underlying asset hits a specific price level, while Asian options depend on the average price of the asset over a set period.
- **Multi-Asset Barrier Options**: Complex options where the payoff depends on multiple underlying assets and includes a barrier feature that requires assets to hit certain price levels.

Why Quantum?

- **Optimization**: Quantum computing can address complex optimization problems in diverse fields, from *logistics* and *machine learning* to *finance*, by exploring vast solution spaces more efficiently than classical methods.
- **Speedup**
  - Classical Monte Carlo: $O(1/\sqrt{M})$ convergence.
  - Quantum Amplitude Estimation: $O(1/M)$, enabling faster convergence for complex, path-dependent options.
- **Where Quantum Shines?**: Quantum is best for complex options (e.g., barrier, Asian) where speedups significantly reduce computational effort.
- **Limitations**: For simple options (e.g., European), classical methods remain more practical and efficient.

General Methodology

Classical

- Model the option's underlying asset price as a set of random variables $\mathbf{X} = \{X_1, X_2, \ldots, X_N\}$ based on a stochastic process.
- Generate M random paths $\{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_M\}$ from the probability distribution of the stochastic process.
- Compute the option's payoff $f(\mathbf{X}_i)$ for each path and estimate the expected payoff $\mathbb{E}_P[f(\mathbf{X})]$ by averaging over all paths: $\hat{\mathbb{E}}_P[f(\mathbf{X})] = \frac{1}{M}\sum_{i=1}^{M} f(\mathbf{X}_i)$ This estimator converges to $\mathbb{E}_P$ as $M \to \infty$, with a convergence rate of $O(M^{-1/2})$.

Quantum

- Represent the probability distribution $\mathbb{P}$ describing the evolution of random variables $\mathbf{X} = \{X_1, X_2, \ldots, X_N\}$ on the quantum computer.
- Construct the circuit which computes the payoff $f(\mathbf{X})$.
- Calculate the expectation value of the payoff $\mathbb{E}_P[f(\mathbf{X})]$.

Amplitude Estimation: Grover Operator

Amplitude Estimation (AE) provides a quantum method to estimate the expected payoff, offering a quadratic speedup compared to classical Monte Carlo methods. Given a unitary operator $\mathcal{A}$ that acts on a register of $(n+1)$ qubits:

$$\mathcal{A}\left|0\right\rangle_{n+1} = \sqrt{1-a}\left|\psi_0\right\rangle_n \left|0\right\rangle + \sqrt{a}\left|\psi_1\right\rangle_n \left|1\right\rangle \tag{1}$$

where $a \in [0,1]$ is the probability amplitude we aim to estimate, which is the probability of measuring $|1\rangle$ in the last qubit. AE uses an operator

$$\mathcal{Q} = \mathcal{A}S_0\mathcal{A}^\dagger S_{\psi_0},$$

where $S_0 = 1 - 2\left|0\right\rangle\left\langle 0\right|$, $S_{\psi_0} = 1 - 2\left|\psi_0\right\rangle\left|0\right\rangle\left\langle\psi_0\right|\left\langle 0\right|$.

This operator $\mathcal{Q}$ performs a rotation in the two-dimensional space spanned by $|\psi_0\rangle|0\rangle$ and $|\psi_1\rangle|1\rangle$.

Amplitude Estimation: Quantum Phase Estimation

Quantum Phase Estimation (QPE) is applied to approximate eigenvalues of $\mathcal{Q}$, which correspond to $a = \sin^2(\theta_a)$. The estimator $\tilde{a}$ for a is given by:

$$\tilde{a} = \sin^2\left(\frac{y\pi}{M}\right) \in [0, 1] \tag{2}$$

where $y \in \{0, \ldots, M-1\}$ and $M = 2^m$ is the number of quantum samples. The m qubits are initialized to an equal superposition state by Hadamard gates, and are used to control different powers of $\mathcal{Q}$. The accuracy of this estimator is:

$$|a - \tilde{a}| \leq \frac{\pi}{M} + \frac{\pi^2}{M^2} = O\left(\frac{1}{M}\right)$$

This provides a quadratic speedup over classical Monte Carlo, which has convergence $O\left(M^{-1/2}\right)$.

Amplitude Estimation: Iterative Amplitude Estimation

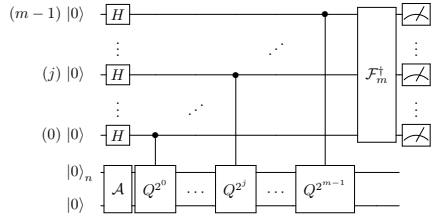


Figure: The quantum circuit of amplitude estimation, where H denotes a Hadamard gate and $\mathcal{F}^\dagger$ the inverse QFT.

To reduce qubit requirements and circuit depth, Suzuki *et al.* demonstrated that AE can achieve quadratic speed-up without QPE. This is done by measuring $\mathcal{Q}^k\mathcal{A}\,|0\rangle$ for $k = 2^0, \ldots, 2^{m-1}$ and using maximum likelihood estimation to approximate θ_a, and thus a. The method exploits $\mathcal{Q}^k\mathcal{A}\,|0\rangle_n\,|0\rangle = \cos\left((2k+1)\theta_a\right)|\psi_0\rangle_n\,|0\rangle + \sin\left((2k+1)\theta_a\right)|\psi_1\rangle_n\,|1\rangle$, where $M = 2^m - 1$ applications of $\mathcal{Q}$ with N shots yield an estimation error scaling as $\mathcal{O}(1/(M\sqrt{N}))$, retaining the speed-up.

Distribution Loading: Uncertainty Model

Log-concave probability distributions can be efficiently loaded in a gate-based quantum computer. Therefore, we construct a circuit to load a log-normal random distribution into a quantum state. The distribution is truncated to a specified interval [low, high] and discretized using 2^n grid points, where n is the number of qubits used. The unitary operator corresponding to this circuit implements the transformation:

$$|0\rangle_n \rightarrow |\psi\rangle_n = \sum_{i=0}^{2^n-1} \sqrt{p_i}\,|i\rangle_n\,, \tag{3}$$

where p_i denotes the probabilities corresponding to the truncated and discretized distribution. The index i is mapped to the interval [low, high] using an affine transformation:

$$\{0,\ldots,2^n-1\} \ni i \mapsto \frac{\text{high}-\text{low}}{2^n-1} \times i + \text{low} \in [\text{low}, \text{high}].$$

Computing the Payoff: State Preparation

For the options considered, the possible values S_i represent the underlying asset prices, each with probability p_i. The operator $\mathcal{A}$ prepares the state:

$$\sum_{i=0}^{2^n-1} \sqrt{1-f(S_i)}\sqrt{p_i}\,|S_i\rangle\,|0\rangle + \sum_{i=0}^{2^n-1} \sqrt{f(S_i)}\sqrt{p_i}\,|S_i\rangle\,|1\rangle\,, \tag{4}$$

where the expectation $a = \sum_{i=0}^{2^n-1} f(S_i)p_i = \mathbb{E}[f(S)]$ represents the option's undiscounted price. Once $\mathcal{A}$ is implemented, we can prepare the state:

$$\mathcal{A}\,|0\rangle = \sum_i \sqrt{p_i}\,|i\rangle\,(\cos(f(i))\,|0\rangle + \sin(f(i))\,|1\rangle)\,, \tag{5}$$

where $f(i) = f_1 i + f_0$ is the linear function since the payoff function for the option contracts of interest is piece-wise linear, with each qubit j in $|i\rangle$ acting as a control for Y-rotations with angle $2^j f_1$. The constant term f_0 is added by a rotation of the ancilla without controls. This setup allows us to map integer values i with probabilities p_i to corresponding values of $f(i)$.

Computing the Payoff: Scaled Expectation

To evaluate $\mathbb{E}[f(X)]$, we scale $f(i)$ as:

$$\tilde{f}(i) = 2\frac{f(i) - f_{\min}}{f_{\max} - f_{\min}} - 1, \tag{6}$$

where $f_{\min}$ and $f_{\max}$ are the minimum and maximum values of $f(i)$. We prepare the state:

$$\sum_{i=0}^{2^n-1} \sqrt{p_i}\,|i\rangle \left(\cos\left(c\tilde{f}(i) + \frac{\pi}{4}\right)|0\rangle + \sin\left(c\tilde{f}(i) + \frac{\pi}{4}\right)|1\rangle\right), \tag{7}$$

where $c \in [0, 1]$ is an additional scaling parameter chosen to ensure $\tilde{f}(i) \in [-1, 1]$.

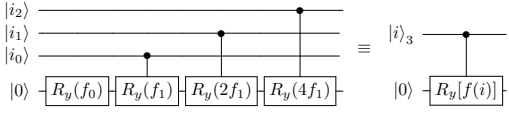


Figure: Quantum circuit for creating the state in Eq. (5). The variable $i = 4i_2 + 2i_1 + i_0 \in \{0, \ldots, 7\}$ is encoded by three qubits as $|i\rangle_3 = |i_2 i_1 i_0\rangle$, with $f(i) = f_1 i + f_0$ expressed as $4f_1 i_2 + 2f_1 i_1 + f_1 i_0 + f_0$. The resulting state is $|i\rangle_3 [\cos(f_1 i + f_0)\,|0\rangle + \sin(f_1 i + f_0)\,|1\rangle]$. The circuit on the right shows a simplified notation.

Computing the Payoff: Probability

The probability of measuring the ancilla qubit in state $|1\rangle$ is:

$$P_1 = \sum_{i=0}^{2^n-1} p_i \sin^2\left(c\tilde{f}(i) + \frac{\pi}{4}\right). \tag{8}$$

Using the approximation $\sin^2\left(c\tilde{f}(i) + \frac{\pi}{4}\right) \approx c\tilde{f}(i) + \frac{1}{2} + \mathcal{O}(c^3\tilde{f}^3(i))$ for small values of $c\tilde{f}(i)$, we get:

$$P_1 \approx c\frac{2\mathbb{E}[f(X)] - f_{\min}}{f_{\max} - f_{\min}} - c + \frac{1}{2}. \tag{9}$$

AE allows us to retrieve P_1 efficiently, enabling the calculation of $\mathbb{E}[f(X)]$. Proper choice of c minimizes estimation error while maintaining AE's quadratic speed-up.

Pricing Options

Vanilla Options: European Call

Vanilla call options are structured so that if the underlying asset price is larger than a fixed value K (the strike price) at the expiration date, the contract pays the difference between the realized price and the strike.

$$f_C(S_T) = \max(0, S_T - K). \tag{10}$$

The implementation uses a comparator, that flips an ancilla qubit from $|0\rangle$ to $|1\rangle$ if $S_T \geq K$, and this ancilla is used to control the linear part of the payoff function.

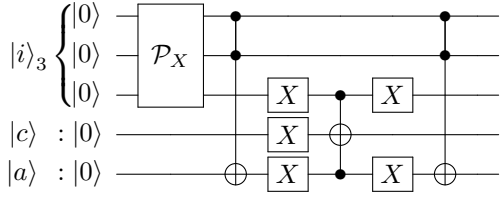


Figure: Quantum circuit that sets the comparator qubit $|c\rangle$ to $|1\rangle$ if $|i\rangle_3$ exceeds strike K. The unitary $\mathcal{P}_X$ loads the probability distribution. Another ancilla qubit $|a\rangle$ is needed, which is uncomputed after the comparison.

Vanilla Options: European Call

The linear part can be approximated by using the fact that $\sin^2(y+\pi/4) \approx y + 1/2$ for small $|y|$. For an approximation rescaling factor $c_{\text{approx}} \in [0,1]$ and $x \in [0,1]$, we consider

$$\sin^2\left(\frac{\pi}{2}\cdot c_{\text{approx}}\cdot\left(x-\frac{1}{2}\right)+\frac{\pi}{4}\right) \approx \frac{\pi}{2}\cdot c_{\text{approx}}\cdot\left(x-\frac{1}{2}\right)+\frac{1}{2}$$

for small values of c_{approx}. An operator can be constructed to act as

$$|x\rangle|0\rangle \to |x\rangle\left(\cos(a\cdot x+b)|0\rangle + \sin(a\cdot x+b)|1\rangle\right), \tag{11}$$

using controlled Y-rotations.

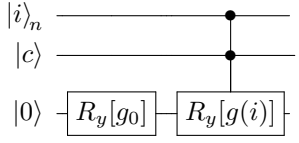


Figure: Circuit that creates the state in Eq. (11). We apply this circuit directly after the comparator circuit shown in Fig. 3. The multi-controlled y-rotation is the gate shown in Fig. 2 controlled by the ancilla qubit $|c\rangle$ that contains the result of the comparison between i and K.

Vanilla Options: Example

We use a price distribution with maturity loaded in a m-qubit register and follow a Black-Scholes-Merton Model (BSM) which implies a log-normal distribution of the asset price at maturity T with probability density,

$$P(S_T) = \frac{1}{S_T \sigma \sqrt{2\pi T}} \exp\left(-\frac{(\ln S_T - \mu)^2}{2\sigma^2 T} \right)$$

where, σ is the volatility of the asset and $\mu = \left(r - 0.5\sigma^2\right) T + \ln(S_0)$, with r the risk-free market rate and S_0 the asset's spot at $t = 0$.

Vanilla Options: Results From the Paper

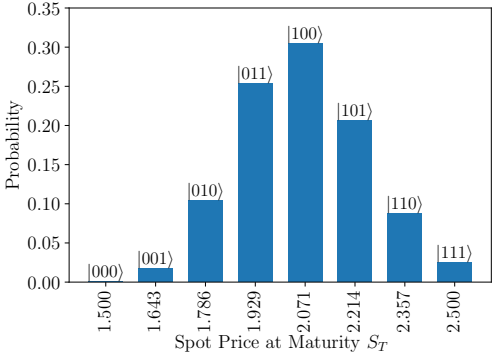


Figure: The authors used $S_0 = 2$, $\sigma = 10\%$, $r = 4\%$, and $T = 300/365$.

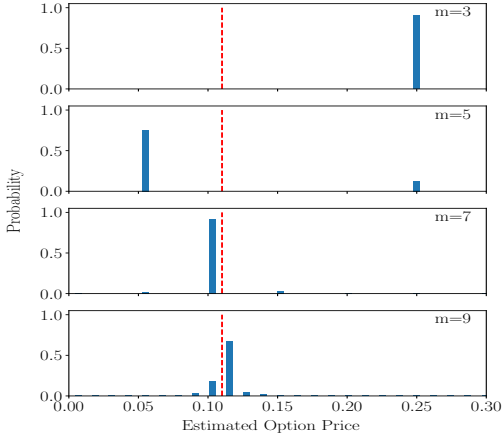


Figure: Results from amplitude estimation on a European call with strike $K = 2.0$ using $m \in \{3, 5, 7, 9\}$ qubits. The red dashed line is the (undiscounted) analytical value from the BSM model.

Vanilla Options: My Results (m = 3)

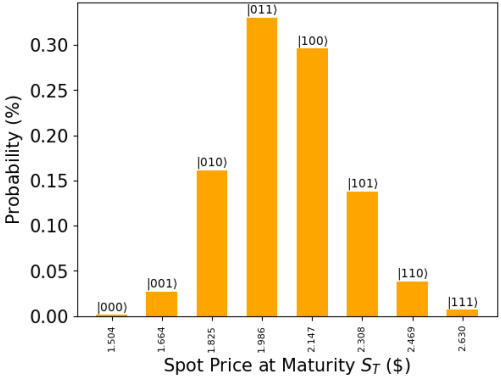


Figure: $S_0 = 2, \sigma = 10\%, r = 4\%$, and $T = 300/365$.

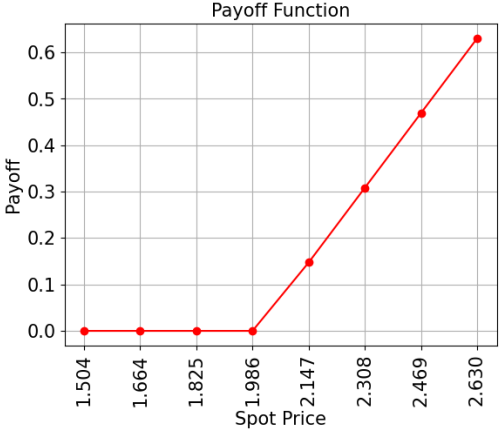


Figure: Payoff function for the distribution defined in Fig. 7

Vanilla Options: My Results (m = 3)

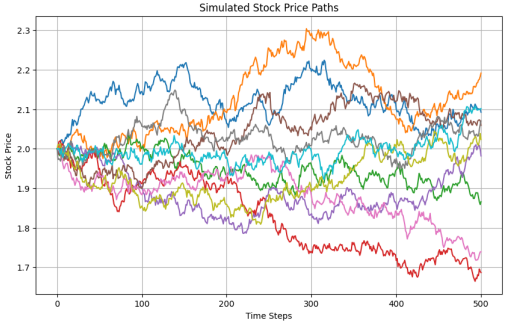


Figure: Monte Carlo simulation of possible stock price paths (plot contains only the first 10 simulations for clarity)

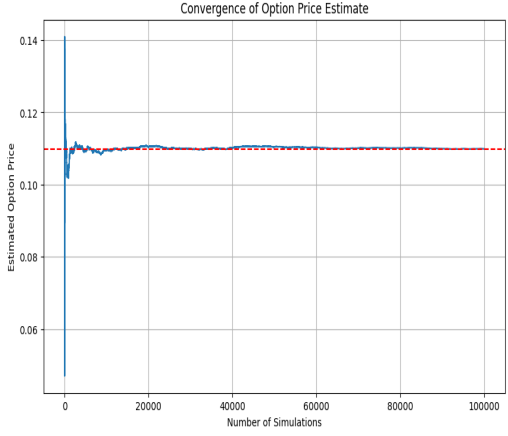


Figure: Convergence of option price estimate.

Vanilla Options: My Results (m = 3)

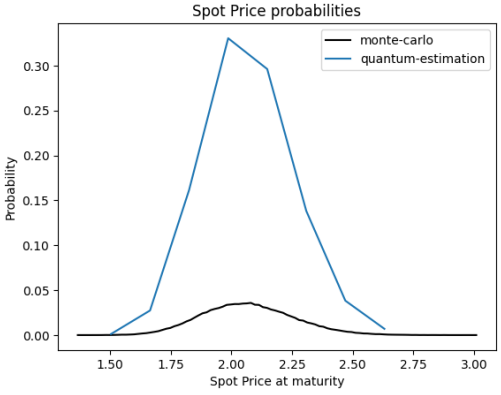


Figure: Comparison between Monte Carlo and Quantum Estimation to load the probability distribution.

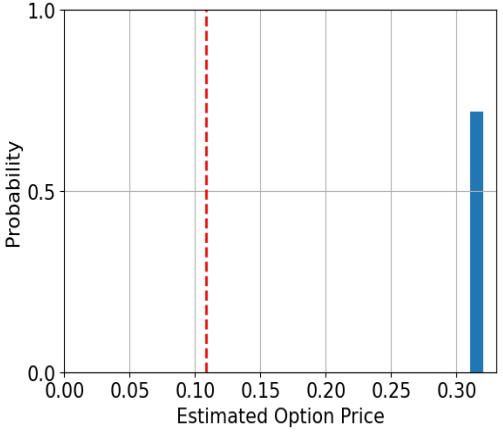


Figure: Results from amplitude estimation on a European call with strike $K = 2.0$ using $m = 3$ qubits. The red dashed line is the (undiscounted) analytical value from the BSM model.

Vanilla Options: My Results

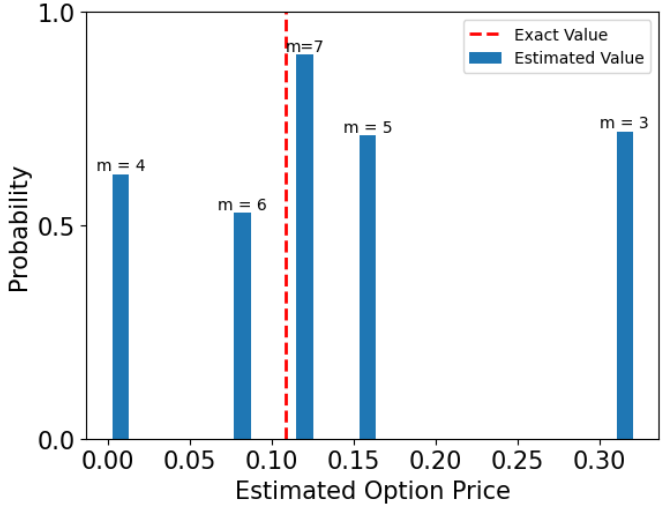


Figure: Results from amplitude estimation on a European call with strike $K = 2.0$ using $m \in \{3, 4, 5, 6, 7\}$ qubits. The red dashed line is the (undiscounted) analytical value from the BSM model.

Asian Options: Payoff

Consider arithmetic average Asian options, which are single-asset, path-dependent options. Their payoff is determined by the underlying asset's price observed at several points throughout the option's life leading up to its expiration date.

$$f(\bar{S}) = \max(0, \bar{S} - K) \tag{12}$$

where K is the strike price, $\bar{S}$ is the arithmetic average of the asset's value over a pre-defined number of points d between 0 and the option maturity T

$$\bar{S} = \frac{1}{d} \sum_{t=1}^{d} S_t. \tag{13}$$

Asian Options: Loading Probability Distribution

The probability distribution of the asset price at time t will again be log-normal with probability density function

$$P(S_t) = \frac{1}{S_t \sigma \sqrt{2\pi \Delta t}} e^{-\frac{(\ln S_t - \mu_t)^2}{2\sigma^2 \Delta t}}, \tag{14}$$

with $\mu_t = \left(r - 0.5\sigma^2\right) \Delta t + \ln(S_{t-1})$ and $\Delta t = T/d$. We can then use the multivariate distribution:

$$P(\vec{S}) = \frac{\exp\left(-\frac{1}{2}(\ln S - \mu)^T \Sigma^{-1} (\ln S - \mu)\right)}{(2\pi)^{d/2} (\det \Sigma)^{1/2} \prod_{i=1}^{d} S_i}, \tag{15}$$

where $\ln S = (\ln S_1, \ln S_2 \ldots, \ln S_d)^T$ and $\mu = (\mu_1, \mu_2 \ldots \mu_d)^T$, where each μ_i is the log-normal distribution parameter for each asset, with $\vec{S}$, a d-dimensional vector of asset prices at time points $[t_1 \ldots t_d]$, instead of distinct underlying prices at maturity T as in the case of Basket Options.

Asian Options

As we are not considering multiple underlying assets that could be correlated, the covariance matrix which is defined as follows:

$$\Sigma = T \begin{bmatrix} \sigma_1^2 & \rho_{12}\sigma_1\sigma_2 & \dots & \rho_{1d}\sigma_1\sigma_d \\ \rho_{21}\sigma_2\sigma_1 & \sigma_2^2 & \dots & \rho_{2d}\sigma_2\sigma_d \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{d1}\sigma_d\sigma_1 & \dots & \dots & \sigma_d^2 \end{bmatrix} \tag{16}$$

is diagonal i.e. $\Sigma = \Delta t[\text{diag}(\sigma^2, ..., \sigma^2)]$, where σ_i the volatility of the ith asset, and $-1 \leq \rho_{ij} \leq 1$ the correlation between assets i and j and T the time to maturity. In this case of asian options, $\sigma_i = \sigma \ \ \forall i \in [d]$ and $\rho_{ij} = 0 \ \ \forall (i,j) \in d \times d$

Asian Options: State Preparation

We now prepare the state

$$|\psi\rangle_n = \sum_{i_1,\ldots,i_d} \sqrt{p_{i_1,\ldots,i_d}}\, |i_1\rangle_{n_1} \otimes \ldots \otimes |i_d\rangle_{n_d}\,, \tag{17}$$

with $n = \sum_j n_j$, where each register represents the asset price at each time step up to maturity. Using the weighted sum operator with equal weights $1/d$, we then calculate the average value of the asset until maturity T, see Eq. (13), into a register $|\bar{S}\rangle$

$$|\underbrace{i_1}_{\Delta t}\underbrace{i_2}_{2\Delta t}\ldots\underbrace{i_d}_{T}\rangle \mapsto |\bar{S}\rangle = |\frac{1}{d}\sum_{t=1}^{d} S_t\rangle\,. \tag{18}$$

Finally, we use the same comparator and rotation circuits that is employed for the basket option to load the payoff of an arithmetic average Asian option into the payoff qubit $|p\rangle$.

Pricing Asian Barrier Spreads

An Asian barrier spread is a combination of three different option types:

- **Asian option**: The payoff depends on the average price over the considered time horizon.
- **Barrier Option**: The payoff is zero if a certain threshold is exceeded at any time within the considered time horizon.
- **(Bull) Spread**: The payoff follows a piecewise linear function (depending on the average price) starting at zero, increasing linear, staying constant.

Suppose strike prices $K_1 < K_2$ and time periods $t = 1, 2$, with corresponding spot prices (S_1, S_2) following a given multivariate distribution (e.g. generated by some stochastic process), and a barrier threshold $B > 0$. The corresponding payoff function is defined as

$$f_P(S_1, S_2) = \begin{cases} \min\left\{\max\left\{\frac{1}{2}(S_1 + S_2) - K_1, 0\right\}, K_2 - K_1\right\}, & \text{if } S_1, S_2 \leq B \\ 0, & \text{otherwise.} \end{cases}$$

Pricing Asian Barrier Spreads: Results

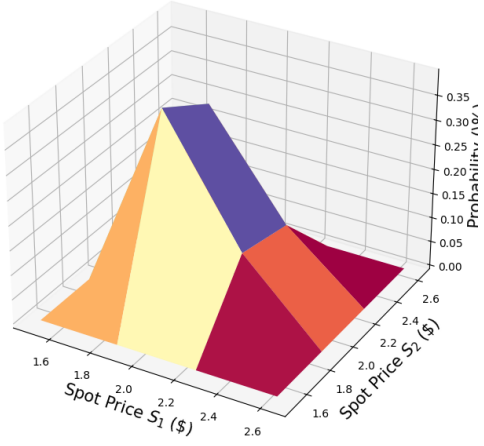


Figure: Probability Distribution Function of the Uncertainty Model.

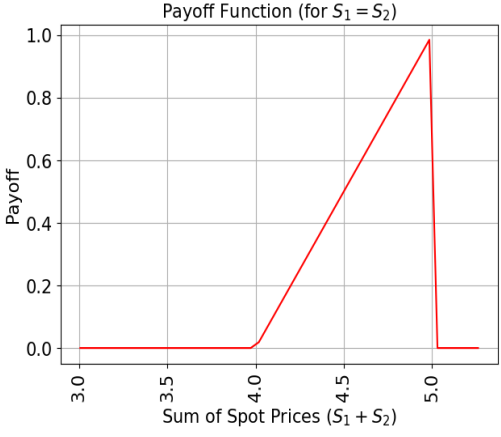


Figure: Exact payoff function.

Pricing Asian Barrier Spreads: Results

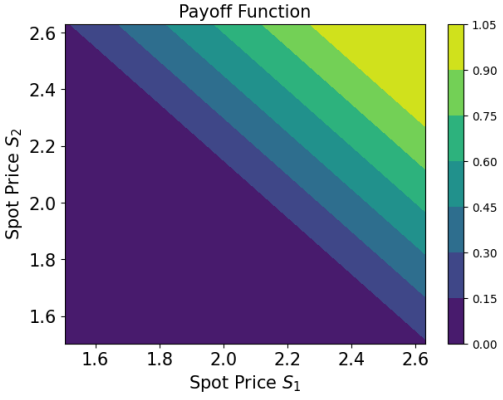


Figure: Payoff function with respect to both time steps ($\Delta t = T/2, 2\Delta t = T$), including barrier ($B = 6.0$)

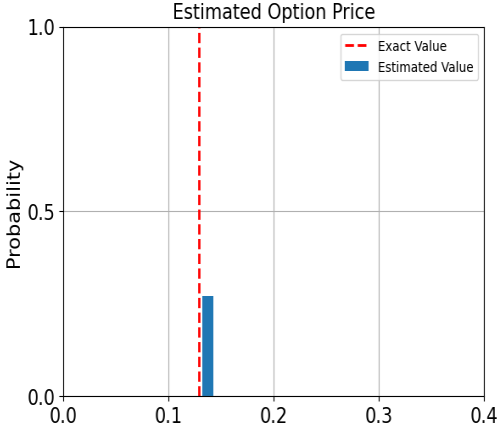


Figure: Results from amplitude estimation on an Asian Barrier Spread with strike $K_1 = 4.0, K_2 = 5.0$. The red dashed line is the exact value.

Conclusion

Feasibility?

Quantum circuits based on standard amplitude estimation are not promising candidates for near-term devices, given that they require extra qubits to control the accuracy of the calculation, and multi-controlled gate operations.

#	Single-qubit	CX	CCX	Depth
$m = 3$	2,091	2,056	90	3,927
$m = 5$	12,768	9,078	378	17,332
$m = 7$	52,275	37,132	1,530	70,916
$m = 9$	210,144	149,290	6,138	285,204

Table: Single-qubit, CNOT, Toffoli gate counts and overall circuit depth required for the full amplitude estimation circuits for European Call Option pricing, as a function of the number of sampling qubits m. These figures assume all-to-all connectivity across qubits.

But, Option Pricing using Iterative Amplitude Estimation is a promising candidate to price options on near term quantum hardware. This, along with results on actual quantum hardware and error mitigation techniques will be discussed in the report.

Summary

- This paper is a foundational paper in the field of quantum finance which provides with a methodology and quantum circuits to price options and option portfolios on gate-based quantum computers.
- It accounts for complex features in exotic options, such as path-dependency with barriers and averages.
- The presented methods are available in the *Qiskit finance module*.
- Pricing options is based on Amplitude Estimation (AE), which offers a quadratic speed-up over classical Monte Carlo simulations.
- Additionally, using AE without Quantum Phase Estimation provides an efficient algorithm which can be realized in the NISQ (Noisy Intermediate-Scale Quantum) era of quantum computation.

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Questions?